Semi-prime factorization – Analysis and improvements on Fermat's Factorization method

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Foreword

This is a work in progress. I need to add a lot more information. However, in this iteration, I hope to compress everything down to as few pages as possible.

No AI was used for this research project. Most of the foundation I uncovered back in 2024 already, but as I had only been studying math for a year, it took me a few additional years to reach the maturity required to make all the right connections. I can however backtrack and explain how I uncovered everything that is described here, from the ground up, with pen and paper, all the way back in 2023 when I started this project.

To do: Add references and closing thoughts and clean everything up.

Chapter I – Fermat's Factorization method

In its most basic form the procedure for Fermat's Factorization method is as follows (where *N* is the semi-prime being factored).

1. Calculate $\sqrt{N}$ in $\mathbb{Z}$
2. Starting from *x* = $\sqrt{N}$, calculate $f(x)=x^2-N$ in a loop, incrementing x.
3. If $f(x)$ evaluates to a square, y^2, calculate the greatest common divisor (*GCD*) on the difference of *x* and *y.*

Taking *N* = 4387 as an example:

Step 1 (Calculate the +/− square root of *N*):

$\sqrt{4387}$ = 66 in $\mathbb{Z}$

Step 2 (Starting from the square root calculate $f(x)=x^2-N$:

66^2 – 4387 = −31 (not a square)
67^2 – 4387 = 102 (not a square)
68^2 – 4387 = 237 (not a square)
69^2 – 4387 = 374 (not a square)
70^2 – 4387 = 513 (not a square)
71^2 – 4387 = 654 (not a square)
72^2 – 4387 = 797 (not a square)
73^2 – 4387 = 942 (not a square)
74^2 – 4387 = 1089 = 33^2 (square)

Step 3 (If *the result is* also a square, calculate the *GCD* on the difference.):

$74^2 = 33^2$ mod 4387

GCD(74 + 33, 4387) = 107
GCD(74 – 33, 4387) = 41

Both the Quadratic Sieve and Number Field Sieve algorithms (to do: insert references) are expansions on this method.

Chapter II – Analysis

<u>a. Binomial expansion</u>

As seen in chapter 1 the goal is the find the following numbers that we can use to take the *GCD* with:

GCD(*x* + *y*, *N*) = *q*
GCD(*x* – *y*, *N*) = *p*

x and y can thus be rewritten as the differences of the factors of *N, p* + *q* or *p* – *q*.
In addition, many more such solutions exist created by (*ap* + *bq*)/2 and (*ap* – *bq*)/2 where *a and b are* multipliers.

If *a* = 13, *b* = 5 and *N* = 4387 (and thus factors *p* = 41 and *q* = 107)

We get:

(13 × 41 + 5 × 107)/2 = 534
(13 × 41 – 5 × 107)/2 = –1

And we can see this creates the following expression when we square 534 modulo *N* (4387 here):

534^2 – 4387 × 13 × 5 = 1^2

This behavior can be explained using binomial expansion.

I will shorten binomial expansion as *BE*($x - y)^e$.
For example *BE*$(x - 74)^2 = x^2 - 148x + 5476$.

Substituting the binomial term x, with for example 33, we get:

BE$(33 - 74)^1 = 33 - 74 = -41$
BE$(33 - 74)^2 = 33^2 - 148 \times 33 + 5476 = -41^2$
BE$(33 - 74)^3 = 33^3 - 222 \times 33^2 + 16428 \times 33 - 405224 = -41^3$

$BE(33 – 74)^4 = 33^4 – 296 × 33^3 + 32856 × 33^2 – 1620896 × 33 + 29986576 = – 41^4$

Conversely we also see the following occurring if we substitute in the factor as root instead (or swapping a binomial term with the binomial value):

$BE(41 – 74)^1 = 41 – 74 = – 33$
$BE(41 – 74)^2 = 41^2 – 148 × 41 + 5476 = – 33^2$
$BE(41 – 74)^3 = 41^3 – 222 × 41^2 + 16428 × 41 – 405224 = – 33^3$
$BE(41 – 74)^4 = 41^4 – 296 × 41^3 + 32856 × 41^2 – 1620896 × 41 + 29986576 = – 33^4$

In addition, if we have even exponents we can substitute the constant by a multiple of N and generate a polynomial p to solve for 0.

$BE(41 – 74)^2 = 41^2 – 148 × 41 + 5476 \rightarrow p = 41^2 – 148 × 41 + 4387 = 0$
$BE(41 – 74)^4 = 41^4 – 296 × 41^3 + 32856 × 41^2 – 1620896 × 41 + 29986576 \rightarrow p = 41^4 – 296 × 41^3 + 32856 × 41^2 – 1620896 × 41 + 4387 × 6565 = 0$

b. The discriminant

Of-course we cannot blindly perform binomial expansion on random numbers. We need some type of measure indicating us how close or far away we are from a solution.
Take for example the following binomial expansion $BE(x – 74)^2 = x^2 – 148x + 5476$ and on this quadratic polynomial p, we substitute the constant with N *(for example N = 4387):*

$p = x^2 – 148x + 4387$

If we now calculate the discriminant of p, $DISC(p) = 148^2 – 4 × 4387 = 66^2$, we see that the result is also square.
And we now have the square relation $148^2 = 66^2 \bmod 4387$, whose GCD(148 – 66, 4387) = 41.

An observation:

From this we can make the following generalization that we can derive the factorization of a semi-prime N, if we find a quadratic polynomial with a constant set to *N, or a multiple there-of*, and whose discriminant yields a square. And from this we can also assume that the discriminant would yield a quadratic residue modulo any prime.

c. Using binomial expansions

The below examples assume $N = 4387$.

Previously we had the following binomial b:

$b = 33 – 74 = – 41$

Next performing binomial expansion to degree 2, if we then substitute the constant by N to create polynomial p, and swap a binomial term with the binomial value (such that $b = 41 – 74$ instead), if p evaluates to 0, we have a valid solution.

$BE(41 – 74)^2 = 41^2 – 148 × 41 + 5476 = 33^2 \rightarrow p = 41^2 – 148 × 41 + 4387 = 0$

For binomial term $y = 74$, only binomial term $x = 33$ will yield a correct binomial value using this setup. Since 74^2 and 33^2 are congruent mod 4387.
Further more calculating the discriminant on polynomial p will yield a square multiple of binomial term x, for example, $DISC(p) = (2 × x)^2$ if we have a correct solution. See previous sub-chapter.

The main issue with this setup when performing binomial expansion using $BE(x – y)^e$, is that when expanding to even exponents, we need to know which multiple of N to substitute with. Because this will determine whether or not a root solution exists, derived from the binomial terms, that will evaluate the Quadratic to 0. Hence using this oracle setup we can restate the factorization problem as finding a multiple of N to substitute the constant value in $BE(x – y)^e$, where e is even, with. Luckily for us, we can calculate the possible multiples of N modulo m (where m is a product of primes). And this is what I propose as an improvement in the next chapter (Chapter 3 – Improvements).

d. An extended example

Let $N = 4387$

Let binomial $b = x – 534$.

Hence $BE(x – 534)^2 = x^2 – 1068x + 534^2$

We now need to substitute the constant in the resulting polynomial with a multiple of N.
For example let multiple $k = 1$.

$k = 1 \rightarrow p = x^2 – 1068x + 4387$

$DISC(p)$ is not a square in $\mathbb{Z}$ hence we know $k = 1$ cannot be a valid solution.

However if $k = 65$:

$k = 65 \rightarrow p = x^2 – 1068x + 4387 × 65$ and $DISC(p) = 2^2$

The discriminant is defined as $DISC(p) = (2 × x)^2$, where x is the binomial term, we can now substitute this into b.

$b(1) = 1 – 534 = -533$ and $p(533) = 0$. Since p evaluates to 0 in $\mathbb{Z}$ we know we have a correct solution and we can take the gcd using our binomial terms:

$GCD(1 + 534, 4387) = 107$
$GCD(534 – 1, 4387) = 41$

In the next chapter we will see how to find these solutions in $\mathbb{Z}/p\mathbb{Z}$

Chapter 3 – Improvements

a. Calculating polynomial residues in $\mathbb{Z}/p\mathbb{Z}$

Let us restrict ourselves to binomial expansions to the second degree.
The way we calculate polynomial residues is that we must consider all possible coefficients in $\mathbb{Z}/p\mathbb{Z}$ whose discriminant evaluates to a quadratic residue.

Here we calculate all possible residues for the quadratic polynomial $ax^2 - bx + Nk$ where $N = 4387$ (you can also calculate them for $ax^2 + bx - Nk$ instead, it just changes the side we are calculating in the expression $x^2 = y^2 \bmod N$). We will ignore the cases where the leading coefficient is 0, as this would downgrade the degree of our polynomial. And in addition we will also ignore the case where $k = 0$ for the sake of brevity.

Mod 3

$k = 1$, $a = 1$ and $b = -1 \rightarrow x = \{ 2 \}$, $k = 1$, $a = 1$ and $b = -2 \rightarrow x = \{ 1 \}$, $k = 1$, $a = 2$ and $b = 0 \rightarrow x = \{ 1, 2 \}$

$k = 2$, $a = 1$ and $b = 0 \rightarrow x = \{ 2, 1 \}$, $k = 2$, $a = 2$ and $b = -1 \rightarrow x = \{ 1 \}$, $k = 2$, $a = 2$ and $b = -2 \rightarrow x = \{ 2 \}$

Mod 5

$k = 1$, $a = 1$ and $b = -2 \rightarrow x = \{ 4, 3 \}$, $k = 1$, $a = 1$ and $b = -3 \rightarrow x = \{ 2, 1 \}$, $k = 1$, $a = 2$ and $b = 0 \rightarrow x = \{ 3, 2 \}$,
$k = 1$, $a = 2$ and $b = -1 \rightarrow x = \{ 4 \}$, $k = 1$, $a = 2$ and $b = -4 \rightarrow x = \{ 1 \}$, $k = 1$, $a = 3$ and $b = 0 \rightarrow x = \{ 1, 4 \}$,
$k = 1$, $a = 3$ and $b = -2 \rightarrow x = \{ 2 \}$, $k = 1$, $a = 3$ and $b = -3 \rightarrow x = \{ 3 \}$, $k = 1$, $a = 4$ and $b = -1 \rightarrow x = \{ 3, 1 \}$,
$k = 1$, $a = 4$ and $b = -4 \rightarrow x = \{ 4, 2 \}$

$k = 2$, $a = 1$ and $b = 0 \rightarrow x = \{ 4, 1 \}$, $k = 2$, $a = 1$ and $b = -1 \rightarrow x = \{ 3 \}$, $k = 2$, $a = 1$ and $b = -4 \rightarrow x = \{ 2 \}$,
$k = 2$, $a = 2$ and $b = -1 \rightarrow x = \{ 2, 1 \}$, $k = 2$, $a = 2$ and $b = -4 \rightarrow x = \{ 3, 4 \}$, $k = 2$, $a = 3$ and $b = -2 \rightarrow x = \{ 3, 1 \}$,
$k = 2$, $a = 3$ and $b = -3 \rightarrow x = \{ 4, 2 \}$, $k = 2$, $a = 4$ and $b = 0 \rightarrow x = \{ 2, 3 \}$, $k = 2$, $a = 4$ and $b = -2 \rightarrow x = \{ 4 \}$,
$k = 2$, $a = 4$ and $b = -3 \rightarrow x = \{ 1 \}$

$k = 3$, $a = 1$ and $b = 0 \rightarrow x = \{ 3, 2 \}$, $k = 3$, $a = 1$ and $b = -2 \rightarrow x = \{ 1 \}$, $k = 3$, $a = 1$ and $b = -3 \rightarrow x = \{ 4 \}$,
$k = 3$, $a = 2$ and $b = -2 \rightarrow x = \{ 2, 4 \}$, $k = 3$, $a = 2$ and $b = -3 \rightarrow x = \{ 1, 3 \}$, $k = 3$, $a = 3$ and $b = -1 \rightarrow x = \{ 3, 4 \}$,
$k = 3$, $a = 3$ and $b = -4 \rightarrow x = \{ 2, 1 \}$, $k = 3$, $a = 4$ and $b = 0 \rightarrow x = \{ 4, 1 \}$, $k = 3$, $a = 4$ and $b = -1 \rightarrow x = \{ 2 \}$,
$k = 3$, $a = 4$ and $b = -4 \rightarrow x = \{ 3 \}$

$k = 4$, $a = 1$ and $b = -1 \rightarrow x = \{ 2, 4 \}$, $k = 4$, $a = 1$ and $b = -4 \rightarrow x = \{ 1, 3 \}$, $k = 4$, $a = 2$ and $b = 0 \rightarrow x = \{ 4, 1 \}$,
$k = 4$, $a = 2$ and $b = -2 \rightarrow x = \{ 3 \}$, $k = 4$, $a = 2$ and $b = -3 \rightarrow x = \{ 2 \}$, $k = 4$, $a = 3$ and $b = 0 \rightarrow x = \{ 2, 3 \}$,
$k = 4$, $a = 3$ and $b = -1 \rightarrow x = \{ 1 \}$, $k = 4$, $a = 3$ and $b = -4 \rightarrow x = \{ 4 \}$, $k = 4$, $a = 4$ and $b = -2 \rightarrow x = \{ 1, 2 \}$,
$k = 4$, $a = 4$ and $b = -3 \rightarrow x = \{ 3, 4 \}$

The problem is, that we can Chinese remainder these together to modulo 15, but then the solution set of possible residues grows quickly (the product of the size of both sets of solutions). So we need to come up with a smart way to generate solutions with this setup.

b. Singular quadratic roots vs non-singular quadratic roots

In the above residue mapping, we will observe that sometimes a set of coefficients will have a single root solution, and sometimes it will have two root solutions.

For example, looking in Mod 5 we see the following two coefficient, root pairings:

$k = 1$, $a = 4$ and $b = -1 \rightarrow x = \{ 3, 1 \}$ Non-singular roots
$k = 1$, $a = 3$ and $b = -2 \rightarrow x = \{ 2 \}$ Singular root

Singular roots indicate that the corresponding binomial term has prime p (5 here) as divisor and that the discriminant is also divided by prime p.

We have calculated the above residues using polynomials of the shape $ax^2 - bx + Nk$.
If we calculate the derivative in $\mathbb{Z}/5\mathbb{Z}$ we get the residue for linear coefficient b in $\mathbb{Z}/5\mathbb{Z}$ corresponding to a quadratic with the signs flipped $a\ ax^2 + bx - Nk$ (the other side of the expression $x^2 = y^2 \bmod N$ that we are trying to solve for).

Calculating the derivative for coefficients with a singular root in $\mathbb{Z}/p\mathbb{Z}$ gives us a derivative that is 0 in $\mathbb{Z}/p\mathbb{Z}$. Since the the derivative calculates the linear coefficient for the other side of the expression $ax^2 - b_0x + Nk = ax^2 + b_1x - Nk$, and since the linear coefficient is constructed by multiplying the binomial term by 2, any divisors of the linear coefficient are hence also divisors of the binomial term.

For example using: $k = 1$, $a = 3$ and $b = -2 \rightarrow x = \{ 2 \}$ we construct polynomial $p = 3 \times 2^2 - 2 \times 2 + 2$ in $\mathbb{Z}/5\mathbb{Z}$.
And we see that the derivative $p' = 2 \times 3 \times 2 - 2 = 0 \bmod 5$ (or in other words 0 in $\mathbb{Z}/5\mathbb{Z}$).

This implies that the binomial term y, in binomial $b = y - x$ equals to 0 in $\mathbb{Z}/5\mathbb{Z}$.

Non-singular roots always yield non-zero residues for the binomial terms in $\mathbb{Z}/p\mathbb{Z}$.

C. p-adic lifting of polynomials in $\mathbb{Z}/p\mathbb{Z}$

Let us say we have the following coefficient and root pairing from our residue mapping:

$k = 4$, $a = 3$ and $b = -1 \rightarrow x = \{ 1 \}$ in $\mathbb{Z}/5\mathbb{Z}$ gives us polynomial $p = 3 \times 1^2 - 1 \times 1 + 4387 \times 4 = 0 \bmod 5$

This gives us all of the following possible solutions in $\mathbb{Z}/25\mathbb{Z}$ where also the derivative yields 0 in $\mathbb{Z}/25\mathbb{Z}$, for the sake of brevity we will focus on these cases only:

$k = 4$, $a = 3$ and $b = -1 \rightarrow x = \{21\}$, $k = 4$, $a = 8$ and $b = -6 \rightarrow x = \{16\}$, $k = 4$, $a = 13$ and $b = -11 \rightarrow x = \{11\}$
$k = 4$, $a = 18$ and $b = -16 \rightarrow x = \{6\}$, $k = 4$, $a = 23$ and $b = -21 \rightarrow x = \{1\}$

$k = 9$, $a = 3$ and $b = -11 \rightarrow x = \{6\}$, $k = 9$, $a = 8$ and $b = -16 \rightarrow x = \{1\}$, $k = 9$, $a = 13$ and $b = -21 \rightarrow x = \{21\}$
$k = 9$, $a = 18$ and $b = -1 \rightarrow x = \{16\}$, $k = 9$, $a = 23$ and $b = -6 \rightarrow x = \{11\}$

$k = 14$, $a = 3$ and $b = -21 \rightarrow x = \{16\}$, $k = 14$, $a = 8$ and $b = -1 \rightarrow x = \{11\}$, $k = 14$, $a = 13$ and $b = -6 \rightarrow x = \{6\}$
$k = 14$, $a = 18$ and $b = -11 \rightarrow x = \{1\}$, $k = 14$, $a = 23$ and $b = -16 \rightarrow x = \{21\}$

k = 19, a = 3 and b = -6 → x = {1}, k = 19, a = 8 and b = -11 → x = {21}, k = 19, a = 13 and b = -16 → x = {16}
k = 19, a = 18 and b = -21 → x = {11}, k = 19, a = 23 and b = -1 → x = {6}

k = 24, a = 3 and b = -16 → x = {11}, k = 24, a = 8 and b = -21 → x = {6}, k = 24, a = 13 and b = -1 → x = {1}
k = 24, a = 18 and b = -6 → x = {21}, k = 24, a = 23 and b = -11 → x = {16}

Each of these solutions map to the same solution in $\mathbb{Z}/5\mathbb{Z}$. We see a total of 25 possible solutions in $\mathbb{Z}/25\mathbb{Z}$ generated from a single solution in $\mathbb{Z}/5\mathbb{Z}$. I refer to p-adic lifting, but usually it is used in the context of lifting roots of polynomials, but in the context that I use this term, it is also used to lift coefficient residues in $\mathbb{Z}/p\mathbb{Z}$ p-adically. So please keep that in mind. I tend to create these abstractions in my head and the names I give these abstract concepts may not be strictly correct in formal mathematical terms. But to me it would seem, that whether we are lifting root residues or coefficient residues, the underlying mechanism are very similar and connected.

D. Converting singular roots to non-singular roots and visa versa

Because we are searching for a two squares such that $x^2 = y^2 \bmod N$, and since both x and y can also be represented as binomials which can be expanded to the second degree, we can substitute the expression $x^2 = y^2 \bmod N$ with the more elaborate: $ax^2 - b_0x + Nk = ax^2 + b_1x - Nk$.

This also means that we can convert singular roots to non-singular roots, simply by transferring a root to the other side of the equation in $\mathbb{Z}/p\mathbb{Z}$. As long as either b_0 or b_1 does not have p as divisor.

For example $1x^2 + 66x - 4387 = 0 \bmod 37$ has a singular root, while the converse $1x^2 - 0x + 4387 = 0 \bmod 37$ has two roots.

$b_1 = 66 \rightarrow 1 \times 4^2 + 66 \times 4 - 4387 = 0 \bmod 37$

$b_0 = 0 \rightarrow 1 \times 4^2 - 0 \times 33 + 4387 = 0 \bmod 37$
$b_0 = 0 \rightarrow 1 \times 33^2 - 0 \times 33 + 4387 = 0 \bmod 37$

But if we flip the sign on the linear coefficient b_1, then the singular root becomes the other non-singular root for b_0 (we must also swap a and k, but both are 1 here):

$b_1 = -66 \rightarrow 1 \times 33^2 - 66 \times 33 - 4387 = 0 \bmod 37$

Chapter 4 – Constructing a better Quadratic Sieve variant

Traditional Quadratic Sieve algorithms are mainly bottle-necked by the fact that beyond 110 digits the factor base required for successful factorization becomes too large. This is mainly due to the relation between factor base size and the on average number of *B*-smooths required to succeed at the linear algebra step. Self Initializing Quadratic Sieve has some improvements, the biggest one being the speed it up gets by guaranteeing the factorization of +/- half of the *B*-smooth candidates being generated. And it also slightly improves our chances of succeeding at linear algebra earlier, as there will be more overlap between factors. SIQS is somewhat the cutting edge of Quadratic Sieve Variants. The next logical improvement would be to achieve more overlap between factors to make the chance of succeeding at the linear algebra step less dependent to the factor base size. For example, if we have a factor base size of 10000 prime numbers and are able to succeed with only 100 *B*-smooths. A new variant like this, would likely also out perform the number field sieve algorithm and give us a real attempt at RSA-1024.

We can use all of the above findings to construct a superior Quadratic Sieve variant.
Our algorithm outline will look like this:

1. Using p-adic lifting, we can calculate root residues for square moduli and perform sieving the same way standard SIQS would (with our method of calculating residues instead), with every *B*-smooth candidate being divisible by at-least the modulus. In this initial phase, the square modulus should ideally have a bit-length close to half the bit-length of semi-prime *N*.

2. Once a *B*-smooth is found by sieving with the square modulus generated in step 1. We divide the *B*-smooth by the square modulus and any other square factors that it may contain and set what remains as a new modulus to go looking for new *B*-smooths. Rather then sieving the roots, we now also use the coefficients and multiples of *N*, as this gives us control over the size of *B*-smooth candidates generated and we want to keep any new factors being introduced as few as possible.

3. Because we will have an *B*-smooth from step one, which without squares will roughly be the size of half the bit-length of *N*, and because this initial *B*-smooth cancels out half of any other *B*-smooths found in step 2, we have now achieved a reduction by half for the required number of *B*-smooths that have to be found in order to be successful during the final linear algebra step (which is the same linear algebra step used in Quadratic Sieve, so I wont reiterate the details of it here).

A much greater reduction in the required amount of B-smooths can also be achieved. We achieve this by sieving in step 3 for *B*-smooths that are smaller then half the bit-length of *N* after dividing by the modulus. The smaller these are, the fewer *B*-smooths we will need. The easiest way to achieve this is by pre-calculating some range of numbers and using coefficients to use these pre-calculated numbers as a offset to the parabola created by the roots. Will demonstrate the details of this below.

An extended example

Let us say semi-prime N = 932774027

During step 1 we find the following *B*-smooth:

$124^2 + 932774027 = 932789403$ using the square 23^2 as modulus
After dividing the *B*-smooth, 932789403, by the modulus we get $932789403 / 23^2 = 1763307 = 3 \times 3 \times 7 \times 13 \times 2153$

Now we move to step 2. We construct a new modulus using the odd exponent factors of the above *B*-smooth.
The modulus thus becomes 195923 ($7 \times 13 \times 2153$).

With this we may find the following *B*-smooth:

$62 \times 4909^2 - 932774027 = 561319395 = 3 \times 5 \times 7 \times 13 \times 191 \times 2153$
Now let us put the factorization of both *B*-smooths atop of each-other ignoring any squares, as these get canceled out during the linear algebra step over $\mathbb{Z}/2\mathbb{Z}$ anyway:

B-smooth from step 1 without squares: $7 \times 13 \times 2153$
B-smooth from step 2 without squares: $3 \times 5 \times 7 \times 13 \times 191 \times 2153$

Now multiplying these together and removing squares we get: $3 \times 5 \times 191 = 2865$

This allows us to cancel out large parts of *B*-smooths, and leaving a much smaller *B*-smooth after taking the product. This effectively reduces the amount of required *B*-smooths to succeed by half, as they effectively become half the size (or even smaller) with this setup.

Previously we generated the following B-smooth during step 2:

$62 \times 4909^2 – 932774027 = 561319395 = 3 \times 5 \times 7 \times 13 \times 191 \times 2153$

One thing to keep in mind is that since 62×4909^2 is not square and since the goal is to find $x^2 – Nk = y^2$, this means we need to add both sides in the matrix for the linear algebra step. However, for the left hand side, we can restrict ourselves to a much smaller factor base (for example, all primes less then 100), hence it does not impact the required number of B-smooth by much.

The most straightforward way to sieve is with the following setup:

$ax^2 – Nk = v$, where v is the polynomial value, or *B*-smooth candidate. And if we can find the right combination of *B*-smooths, whose product is square, then from this we derive y^2.

So as long as a factors over the factor base, then by extension ax^2 also factors over the factor base, since we can functionally ignore any squares regardless of factorization.

The trick to generating a small polynomial value, or B-smooth candidate, using $ax^2 – Nk$ is that we must find some ax^2 whose bit-length is close to Nk. This is quite convenient, because this means we can use a to add, or remove, bits to a root and get it closer to Nk in order to reduce the size of the polynomial value.

This has its limitations, since this coefficient a, would still be a multiplier to the square of a fairly large root if we use a modulus that is roughly half the bit-length of N.
However in step 2, we can instead create different moduli that are ¼th the bit-length of N. This now means that when sieving $ax^2 – Nk$, coefficient a is a multiplier to a much smaller square. This does mean that a now must be a much greater number, roughly the size of the square root of N. However, we can sieve in advance numbers around the square root of N and find cases where they factorize over a small factor base. And these numbers can then also be re-used for any other semi-prime N of that bit-size we try to factorize in the future.

In addition, if the root also factorizes over the factor base, we could even use the linear coefficient for much great control, sieving $ax^2 + bx – Nk$ instead.. Then all of this we can extend even further by instead of trying sieivng $ax^2 – Nk = v$ we go looking for $ax^4 – Nk = v$, using k to add bits to N when needed and using the other coefficients aswell, for example $ax^4 + bx^3 – Nk = v$

To do: Finish the code first, then show an extended example using everything said here.